Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 04:
SQL Joins & Exercises

Digital for Executive

Implicit Join
(Access Multiple Tables)

Implicit JOIN

Specify 2 tables in the FROM clause:

```
select * from orderdetails, orders;
```

The result is a full join (or Cartesian join):

- Every row in the first table is joined with every row in the second table
- The result set will have more rows than in both tables

Implicit JOIN

- Use additional operands to limit the result set:

```
SELECT
    *
FROM orderdetails,orders
WHERE orderdetails.orderNumber = orders.orderNumber;
```

- Use shorter aliases for table names:

```
SELECT
    *
FROM orderdetails D, orders O
WHERE D.orderNumber = O.orderNumber;
```

Implicit JOIN

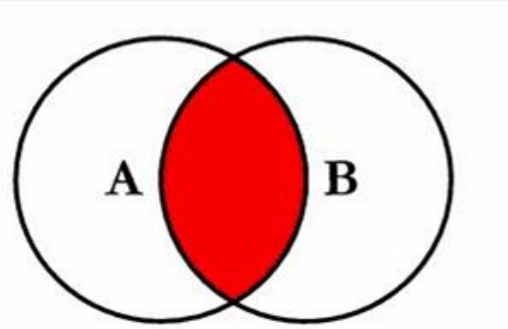
- See **orderNumber, productCode, orderDate** from orderdetails and orders tables. Column names in the select clause can be pre-fixed by aliases:

```
SELECT
   D.orderNumber, D.productCode, O.orderDate
FROM
   orderdetails D,
   orders O
WHERE
   D.orderNumber = O.orderNumber;
```

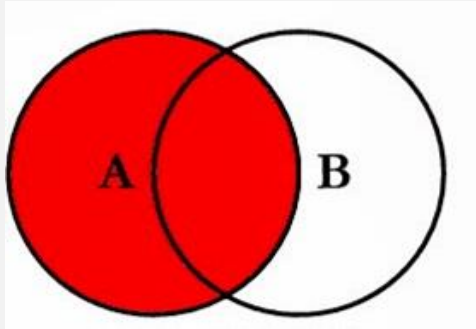
orderNumber	productCode	orderDate
10100	S18_1749	2003-01-06
10100	S18_2248	2003-01-06
10100	S18_4409	2003-01-06
10100	S24_3969	2003-01-06
10101	S18_2325	2003-01-09
10101	S18_2795	2003-01-09
10101	S24_1937	2003-01-09
10101	S24_2022	2003-01-09
10102	S18_1342	2003-01-10
10102	S18_1367	2003-01-10
10103	S10_1949	2003-01-29
10103	S10_4962	2003-01-29
10103	S12_1666	2003-01-29
10103	S18_1097	2003-01-29

JOIN Table

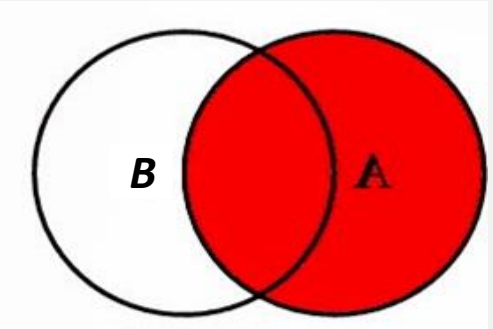
- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.
- Here are the different types of the JOINs in SQL:
  1. **(INNER) JOIN**: Returns records that have matching values in both tables
  2. **LEFT (OUTER) JOIN**: Returns all records from the left table, and the matched records from the right table
  3. **RIGHT (OUTER) JOIN**: Returns all records from the right table, and the matched records from the left table
  4. **FULL (OUTER) JOIN**: Returns all records when there is a match in either left or right table



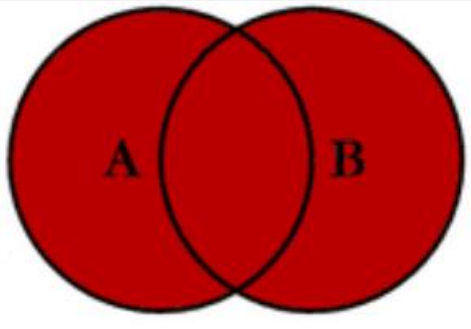
Inner Join



Left Join



Right Join



Full Outer Join

Training Database File

1. Open “training_db.sql” file using any text editor
2. In MySQL Workbench, Run the script contained in the file
3. Run syntax “use classicmodels;”

Tables can be seen on the left

*you will use this database for practice in the upcoming sessions to understand about join table

DF_1

var_id	var2
a11	1
a12	2
a13	3
a14	5

DF_2

var_id	var3
a12	5
a13	6
a14	7
a15	8

INNER JOIN

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Exemples
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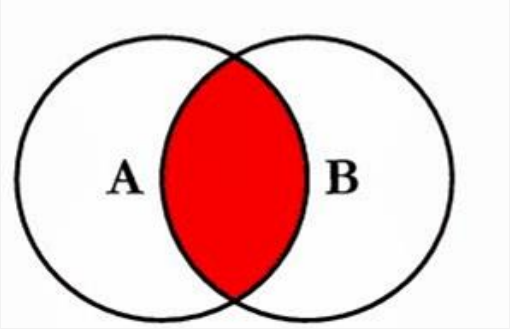
INNER JOIN

The INNER JOIN keyword selects records that have matching values in both tables.

```
SELECT column_name(s)
FROM table1
INNER JOIN table2
ON table1.column_name =
table2.column_name;
```



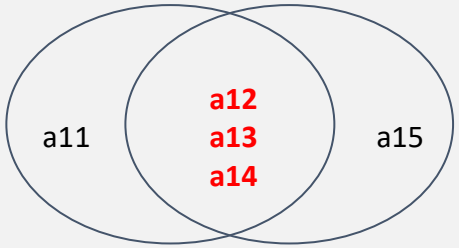
INNER JOIN
Simple Example

DF_1

	var_id	var2
▶	a11	1
	a11	10
	a12	2
	a12	4
	a13	3
	a14	5

DF_2

	var_id	var3
▶	a12	5
	a13	6
	a14	7
	a15	8



query:

```
SELECT
   DF_1.var_id, DF_1.var2, df_2.var3
FROM
   DF_1
      JOIN
   DF_2 ON DF_1.var_id = DF_2.var_id;
```

Result

	var_id	var2	var3
▶	a12	2	5
	a12	4	5
	a13	3	6
	a14	5	7

INNER JOIN

- **customers** INNER JOIN with **orders** based on **customerNumber**. Show only **customerNumber, customerName,orderDate, status**

customers

customerNumber	customerName
103	Atelier graphique
112	Signal Gift Stores
114	Australian Collectors, Co.
119	La Rochelle Gifts
121	Baane Mini Imports
124	Mini Gifts Distributors Ltd.
125	Havel & Zbyszek Co
128	Blauer See Auto, Co.
129	Mini Wheels Co.
131	Land of Toys Inc.
141	Euro+ Shopping Channel
144	Volvo Model Replicas, Co
145	Danish Wholesale Imports
146	Saveley & Henriot, Co.
148	Dragon Souveniers, Ltd.

orders

customerNumber	orderDate	status
363	2003-01-06	Shipped
128	2003-01-09	Shipped
181	2003-01-10	Shipped
121	2003-01-29	Shipped
141	2003-01-31	Shipped
145	2003-02-11	Shipped
278	2003-02-17	Shipped
131	2003-02-24	Shipped
385	2003-03-03	Shipped
486	2003-03-10	Shipped
187	2003-03-18	Shipped
129	2003-03-25	Shipped
144	2003-03-24	Shipped
124	2003-03-26	Shipped
172	2003-04-01	Shipped

INNER JOIN

```
SELECT
    C.customerNumber,
    C.customerName,
    O.orderDate,
    O.status
FROM
    customers C
        INNER JOIN
    orders O ON C.customerNumber = O.customerNumber;
```

Join result, should be 326 rows

customerNumber	customerName	orderDate	status
363	Online Diecast Creations Co.	2003-01-06	Shipped
128	Blauer See Auto, Co.	2003-01-09	Shipped
181	Vitachrome Inc.	2003-01-10	Shipped
121	Baane Mini Imports	2003-01-29	Shipped
141	Euro+ Shopping Channel	2003-01-31	Shipped
145	Danish Wholesale Imports	2003-02-11	Shipped
278	Rovelli Gifts	2003-02-17	Shipped
131	Land of Toys Inc.	2003-02-24	Shipped
385	Cruz & Sons Co.	2003-03-03	Shipped
486	Motor Mint Distributors Inc.	2003-03-10	Shipped
187	AV Stores, Co.	2003-03-18	Shipped
129	Mini Wheels Co.	2003-03-25	Shipped
144	Volvo Model Replicas, Co	2003-03-24	Shipped
124	Mini Gifts Distributors Ltd.	2003-03-26	Shipped
172	La Corne D'abondance, Co.	2003-04-01	Shipped

LEFT JOIN

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Exemples
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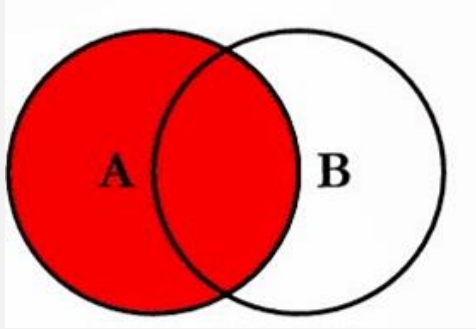
LEFT JOIN

- The LEFT JOIN keyword returns all records from the left table (table1), and the matched records from the right table (table2). The result is NULL from the right side, if there is no match.
- In some databases LEFT JOIN is called LEFT OUTER JOIN.

```
SELECT column_name(s)
FROM table1
LEFT JOIN table2
ON table1.column_name =
table2.column_name;
```



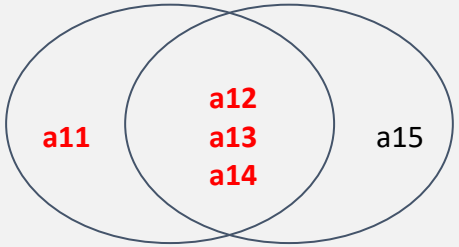
LEFT JOIN
Simple Example

DF_1

var_id	var2
a11	1
a11	10
a12	2
a12	4
a13	3
a14	5

DF_2

var_id	var3
a12	5
a13	6
a14	7
a15	8



query:

```
SELECT
   DF_1.var_id, DF_1.var2, df_2.var3
FROM
   DF_1
      LEFT JOIN
   DF_2 ON DF_1.var_id = DF_2.var_id;
```

Result

var_id	var2	var3
a11	1	NULL
a11	10	NULL
a12	2	5
a12	4	5
a13	3	6
a14	5	7

LEFT JOIN

- **customers** Left JOIN with **orders** based on **customerNumber**. Show only **customerNumber, customerName,orderDate, status**

customers

customerNumber	customerName
103	Atelier graphique
112	Signal Gift Stores
114	Australian Collectors, Co.
119	La Rochelle Gifts
121	Baane Mini Imports
124	Mini Gifts Distributors Ltd.
125	Havel & Zbyszek Co
128	Blauer See Auto, Co.
129	Mini Wheels Co.
131	Land of Toys Inc.
141	Euro+ Shopping Channel
144	Volvo Model Replicas, Co
145	Danish Wholesale Imports
146	Saveley & Henriot, Co.
148	Dragon Souveniers, Ltd.

orders

customerNumber	orderDate	status
363	2003-01-06	Shipped
128	2003-01-09	Shipped
181	2003-01-10	Shipped
121	2003-01-29	Shipped
141	2003-01-31	Shipped
145	2003-02-11	Shipped
278	2003-02-17	Shipped
131	2003-02-24	Shipped
385	2003-03-03	Shipped
486	2003-03-10	Shipped
187	2003-03-18	Shipped
129	2003-03-25	Shipped
144	2003-03-24	Shipped
124	2003-03-26	Shipped
172	2003-04-01	Shipped

LEFT JOIN

```
SELECT
    C.customerNumber,
    C.customerName,
    O.orderDate,
    O.status
FROM
    customers C
        Left JOIN
    orders O ON C.customerNumber = O.customerNumber;
```

Join result, should be 350 rows, several null values from orderDate and status

customerNumber	customerName	orderDate	status
223	Natürlich Autos	NULL	NULL
227	Heintze Collectables	2003-10-17	Shipped
227	Heintze Collectables	2004-10-22	Shipped
233	Québec Home Shopping ...	2003-11-05	Shipped
233	Québec Home Shopping ...	2004-06-17	Shipped
233	Québec Home Shopping ...	2005-05-01	Shipped
237	ANG Resellers	NULL	NULL
239	Collectable Mini Designs ...	2004-02-19	Shipped
239	Collectable Mini Designs ...	2004-02-26	Shipped
240	giftsbymail.co.uk	2004-03-20	Shipped
240	giftsbymail.co.uk	2004-11-01	Shipped
242	Alpha Cognac	2003-07-04	Shipped
242	Alpha Cognac	2003-11-08	Shipped
242	Alpha Cognac	2005-03-28	Shipped
247	Messner Shopping Network	NULL	NULL

RIGHT JOIN

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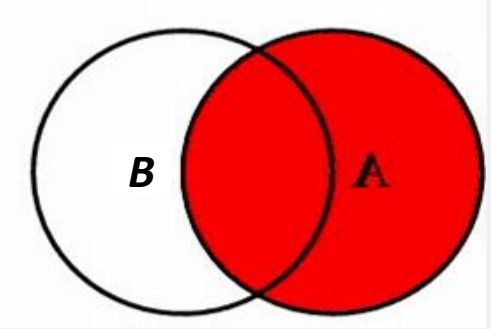
RIGHT JOIN

- The RIGHT JOIN keyword returns all records from the right table (table2), and the matched records from the left table (table1). The result is NULL from the left side, when there is no match.
- In some databases RIGHT JOIN is called RIGHT OUTER JOIN.

```
SELECT column_name(s)
FROM table1
RIGHT JOIN table2
ON table1.column_name =
table2.column_name;
```



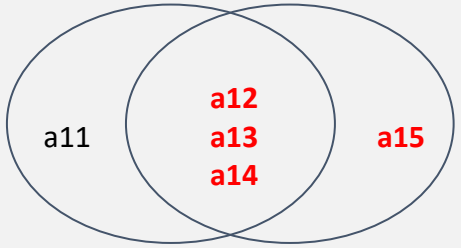
RIGHT JOIN
Simple Example

DF_1

var_id	var2
a11	1
a11	10
a12	2
a12	4
a13	3
a14	5

DF_2

var_id	var3
a12	5
a13	6
a14	7
a15	8



query:

```
SELECT
   DF_2.var_id, DF_1.var2, df_2.var3
FROM
   DF_1
      RIGHT JOIN
   DF_2 ON DF_1.var_id = DF_2.var_id;
```

Result

var_id	var2	var3
a12	2	5
a12	4	5
a13	3	6
a14	5	7
a15	NULL	8

RIGHT JOIN

- **customers** Right JOIN with **orders** based on **customerNumber.** Show only **customerNumber, customerName,orderDate, status**

customers

customerNumber	customerName
103	Atelier graphique
112	Signal Gift Stores
114	Australian Collectors, Co.
119	La Rochelle Gifts
121	Baane Mini Imports
124	Mini Gifts Distributors Ltd.
125	Havel & Zbyszek Co
128	Blauer See Auto, Co.
129	Mini Wheels Co.
131	Land of Toys Inc.
141	Euro+ Shopping Channel
144	Volvo Model Replicas, Co
145	Danish Wholesale Imports
146	Saveley & Henriot, Co.
148	Dragon Souveniers, Ltd.

orders

customerNumber	orderDate	status
363	2003-01-06	Shipped
128	2003-01-09	Shipped
181	2003-01-10	Shipped
121	2003-01-29	Shipped
141	2003-01-31	Shipped
145	2003-02-11	Shipped
278	2003-02-17	Shipped
131	2003-02-24	Shipped
385	2003-03-03	Shipped
486	2003-03-10	Shipped
187	2003-03-18	Shipped
129	2003-03-25	Shipped
144	2003-03-24	Shipped
124	2003-03-26	Shipped
172	2003-04-01	Shipped

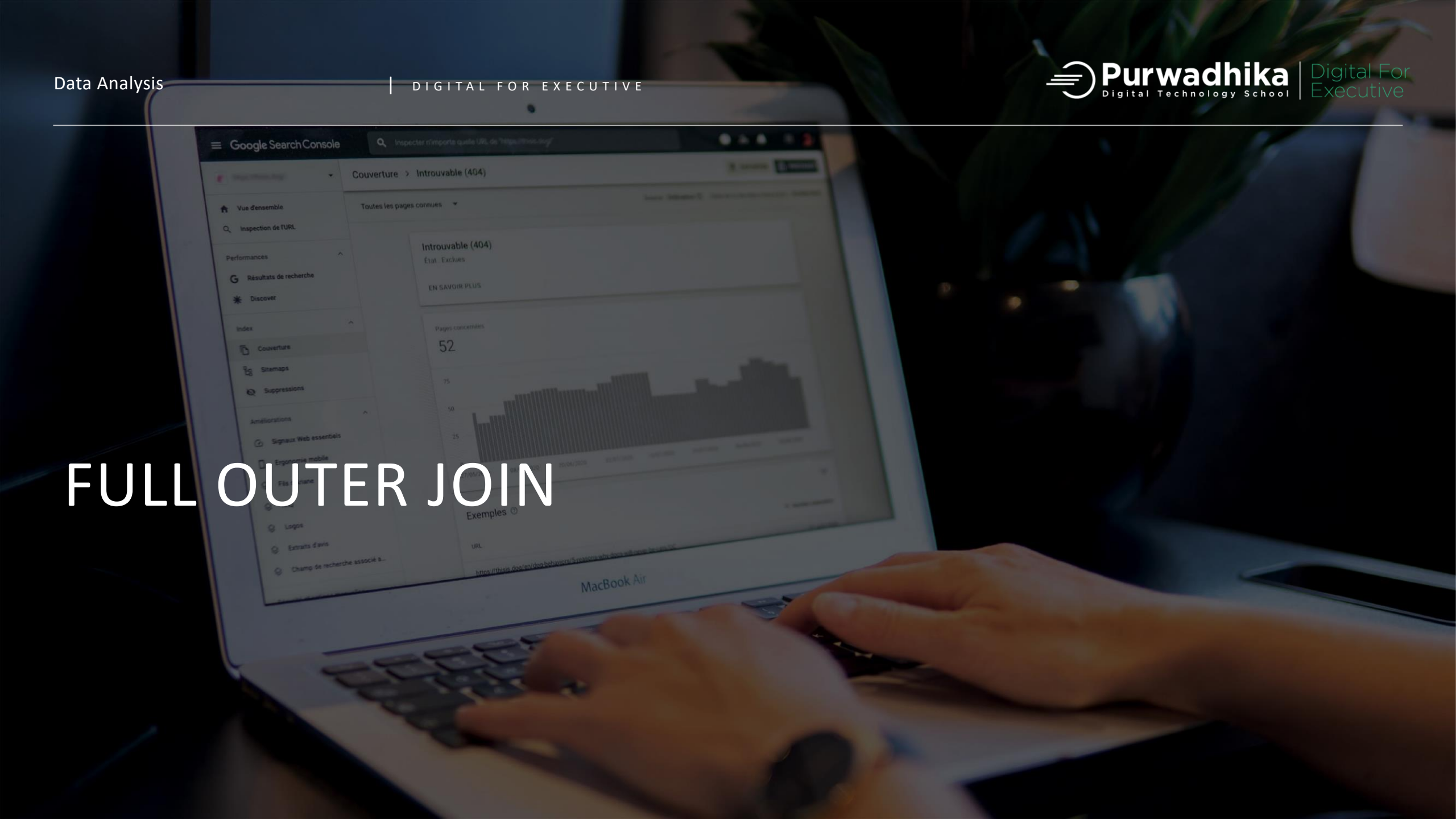
RIGHT JOIN

```
SELECT
    C.customerNumber,
    C.customerName,
    O.orderDate,
    O.status
FROM
    orders O
        Right JOIN
    customers C ON O.customerNumber = C.customerNumber;
```

Join result, should be 350 rows, several null values from orderDate and status

customerNumber	customerName	orderDate	status
223	Natürlich Autos	NULL	NULL
227	Heintze Collectables	2003-10-17	Shipped
227	Heintze Collectables	2004-10-22	Shipped
233	Québec Home Shopping ...	2003-11-05	Shipped
233	Québec Home Shopping ...	2004-06-17	Shipped
233	Québec Home Shopping ...	2005-05-01	Shipped
237	ANG Resellers	NULL	NULL
239	Collectable Mini Designs ...	2004-02-19	Shipped
239	Collectable Mini Designs ...	2004-02-26	Shipped
240	giftsbymail.co.uk	2004-03-20	Shipped
240	giftsbymail.co.uk	2004-11-01	Shipped
242	Alpha Cognac	2003-07-04	Shipped
242	Alpha Cognac	2003-11-08	Shipped
242	Alpha Cognac	2005-03-28	Shipped
247	Messner Shopping Network	NULL	NULL

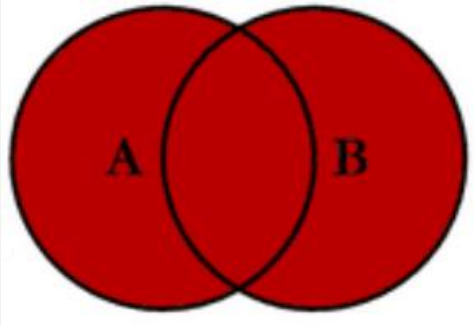
FULL OUTER JOIN



FULL OUTER JOIN

- We don't have FULL JOINS on MySQL, but we can sure emulate them.

```
SELECT column_name(s)
FROM table1
LEFT JOIN table2
ON table1.column_name = table2.column_name
UNION
SELECT column_name(s)
FROM table1
RIGHT JOIN table2
ON table1.column_name = table2.column_name
```



SQL Exercises

Guide

- Database untuk latihan adalah database bernama “sakila”. Jika belum memiliki database tersebut, download terlebih dahulu databasenya pada laman https://dev.mysql.com/doc/index-other.html
- Jawaban dapat dituliskan pada MySQL Workbench

1st Challenge

- Gunakan database “Sakila”!
- Dari tabel “payment”, tampilkan 10 baris data customer id, rental id, amount, dan payment date!

customer_id	rental_id	amount	payment_date
1	76	2.99	2005-05-25 11:30:37
1	573	0.99	2005-05-28 10:35:23
1	1185	5.99	2005-06-15 00:54:12
1	1422	0.99	2005-06-15 18:02:53
1	1476	9.99	2005-06-15 21:08:46
1	1725	4.99	2005-06-16 15:18:57
1	2308	4.99	2005-06-18 08:41:48
1	2363	0.99	2005-06-18 13:33:59
1	3284	3.99	2005-06-21 06:24:45
1	4526	5.99	2005-07-08 03:17:05

2nd Challenge

- Gunakan database “Sakila”!
- Dari tabel “film”, tampilkan 10 judul film, tahun release, dan durasi rental dimana judul film yang ditampilkan yang dimulai dengan huruf “S”!

title	release_year	rental_duration
SABRINA MIDNIGHT	2006	5
SADDLE ANTITRUST	2006	7
SAGEBRUSH CLUELESS	2006	4
SAINTS BRIDE	2006	5
SALUTE APOLLO	2006	4
SAMURAI LION	2006	5
SANTA PARIS	2006	7
SASSY PACKER	2006	6
SATISFACTION CONFIDENTIAL	2006	3
SATURDAY LAMBS	2006	3

3rd Challenge

- Gunakan database “Sakila”!
- Dari tabel “film”, tampilkan durasi rental, banyaknya film setiap durasi rental, dan rata-rata durasi film!
- Kelompokkan jumlah film dan rata-rata durasi film berdasarkan durasi rentalnya!
- Karena rata-rata durasi film angka desimal, bulatkan 2 angka di belakang koma!

Durasi_Rental	Banyak_Film	Rata_Rata_Durasi_Film
3	203	112.91
4	203	111.23
5	191	116.56
6	212	117.04
7	191	118.83

4th Challenge

- Gunakan database “Sakila”!
- Dari tabel “film”, tampilkan title, durasi film, dan rating yang durasi filmnya lebih dari rata-rata durasi film total!
- Tampilkan 25 data yang diurutkan dari durasi terlama!

title	length	rating
INSTINCT AIRPORT	116	PG
MADIGAN DORADO	116	R
DUCK RACER	116	NC-17
WORDS HUNTER	116	PG
AFFAIR PREJUDICE	117	G
GRAFFITI LOVE	117	PG
MAGIC MALLRATS	117	PG
CHAMBER ITALIAN	117	NC-17
RESURRECTION SILVERADO	117	PG
ORIENT CLOSER	118	R
PATHS CONTROL	118	PG
JAWBREAKER BROOKLYN	118	PG
OTHERS SOUP	118	PG
BACKLASH UNDEFEATED	118	PG-13
PERSONAL LADYBUGS	118	PG-13
ROUGE SQUAD	118	NC-17
FIDELITY DEVIL	118	G
DUMBO LUST	119	NC-17
STRANGERS GRAFFITI	119	R
IDENTITY LOVER	119	PG-13
APOCALYPSE FLAMINGOS	119	R
BUGSY SONG	119	G
GAMES BOWFINGER	119	PG-13
DOLLS RAGE	120	PG-13
CALENDAR GUNFIGHT	120	NC-17

5th Challenge

- Gunakan database “Sakila”!
- Dari tabel “film”, tampilkan rating, Replacement Cost tertinggi, Rental Rate Terendah, dan Rata-Rata Durasi dan kelompokkan berdasarkan rating film!

Rating	Replacement_Cost_Tertinggi	Rental_Rate_Terendah	Rata_Rata_Durasi
G	29.99	0.99	111.0506
PG	29.99	0.99	112.0052
PG-13	29.99	0.99	120.4439
R	29.99	0.99	118.6615
NC-17	29.99	0.99	113.2286

6th Challenge

- Gunakan database “Sakila”!
- Tampilkan 15 daftar film yang memiliki huruf “K” pada akhir pada title, lalu tampilkan title, durasi, serta Bahasa pada film!
- Sebagai catatan, lakukan join terlebih dahulu dari tabel “film” dan tabel “language”!

Judul	Durasi	Bahasa_Film
AIRPORT POLLOCK	54	English
BEAST HUNCHBACK	89	English
BULL SHAWSHANK	125	English
CANYON STOCK	85	English
CHILL LUCK	142	English
CHITTY LOCK	107	English
CHOCOLATE DUCK	132	English
CITIZEN SHREK	165	English
CROWDS TELEMARK	112	English
DAZED PUNK	120	English
DESTINATION JERK	76	English
FANTASIA PARK	131	English
FLYING HOOK	69	English
GRIT CLOCKWORK	137	English
HANDICAP BOONDOCK	108	English

7th Challenge

- Gunakan database “Sakila”!
- Tampilkan Judul Film (dari tabel “film”), First Name (dari tabel “actor”), dan Last Name (Dari tabel “actor”) dari actor yang memiliki “actor_id” = 14!
- Sebagai catatan, lakukan join table terlebih dahulu antara tabel “film”, “film_actor”, dan “actor”!

Judul_Film	First_Name	Last_Name
CLASH FREDDY	VIVIEN	BERGEN
CRANES RESERVOIR	VIVIEN	BERGEN
DIRTY ACE	VIVIEN	BERGEN
DONNIE ALLEY	VIVIEN	BERGEN
DRIFTER COMMANDMENTS	VIVIEN	BERGEN
DRIVING POLISH	VIVIEN	BERGEN
DRUMS DYNAMITE	VIVIEN	BERGEN
ENEMY ODDS	VIVIEN	BERGEN
EXCITEMENT EVE	VIVIEN	BERGEN
GORGEOUS BINGO	VIVIEN	BERGEN
HIGH ENCINO	VIVIEN	BERGEN
HILLS NEIGHBORS	VIVIEN	BERGEN
HOBBIT ALIEN	VIVIEN	BERGEN
IMPACT ALADDIN	VIVIEN	BERGEN
ITALIAN AFRICAN	VIVIEN	BERGEN
JAPANESE RUN	VIVIEN	BERGEN
KENTUCKIAN GIANT	VIVIEN	BERGEN
LOVELY JINGLE	VIVIEN	BERGEN
LOVER TRUMAN	VIVIEN	BERGEN
MUSSOLINI SPOILERS	VIVIEN	BERGEN
POLISH BROOKLYN	VIVIEN	BERGEN
SALUTE APOLLO	VIVIEN	BERGEN
SATURDAY LAMBS	VIVIEN	BERGEN
STOCK GLASS	VIVIEN	BERGEN
STREETCAR INTENTIONS	VIVIEN	BERGEN
TIGHTS DAWN	VIVIEN	BERGEN
TRAP GUYS	VIVIEN	BERGEN
TYCOON GATHERING	VIVIEN	BERGEN
VOICE PEACH	VIVIEN	BERGEN
WESTWARD SEABISCUIT	VIVIEN	BERGEN

8th Challenge

- Gunakan database “Sakila”!
- Dari tabel "city“, tampilkan city dan country id!
- Tampilkan hanya city atau kota yang namanya ada huruf "d" di posisi mana pun dan diakhiri huruf “a”!
- Tampilkan 15 data yang diurutkan berdasarkan city-nya secara ascending!

city	country_id
Adana	97
Addis Abeba	31
Alessandria	49
Alvorada	15
Aparecida de Goinia	15
Avellaneda	6
Bamenda	19
Crdoba	6
Dhaka	12
Guadalajara	60
Haldia	44
Hodeida	107
Jedda	82
Juiz de Fora	15
Kaduna	69

9th Challenge

- Gunakan database “Sakila”!
- Tampilkan nama genre (name dari tabel “category”) dan jumlah banyaknya film di setiap genrenya!
- Lakukan join terlebih dahulu antara tabel “film”, “film_category”, dan “category”!
- Urutkan berdasarkan jumlah film di setiap kategorinya secara ascending!

Genre	Banyak_Film
Music	51
Horror	56
Travel	57
Classics	57
Comedy	58
Children	60
Sci-Fi	61
Games	61
Drama	62
New	63
Action	64
Animation	66
Documentary	68
Family	69
Foreign	73
Sports	74

10th Challenge

- Gunakan database “Sakila”!
- Dari tabel “film”, tampilkan title, description, length, serta rating!
- Tampilkan 10 judul film yang diakhiri huruf ‘h’ dan durasinya di atas durasi rata-rata!
- Urutkan berdasarkan judulnya secara ascending!

title	description	length	rating
CHICAGO NORTH	A Fateful Yarn of a Mad Cow And a Waitress w...	185	PG-13
DADDY PITTSBURGH	A Epic Story of a A Shark And a Student who m...	161	G
DOUBLE WRATH	A Thoughtful Yarn of a Womanizer And a Dog w...	177	R
DOUBTFIRE LABYRINTH	A Intrepid Panorama of a Butler And a Compose...	154	R
DRIVING POLISH	A Action-Packed Yarn of a Feminist And a Techn...	175	NC-17
EMPIRE MALKOVICH	A Amazing Story of a Feminist And a Cat who m...	177	G
GREASE YOUTH	A Emotional Panorama of a Secret Agent And a ...	135	G
HAROLD FRENCH	A Stunning Saga of a Sumo Wrestler And a Stud...	168	NC-17
HOMICIDE PEACH	A Astounding Documentary of a Hunter And a B...	141	PG-13
JADE BUNCH	A Insightful Panorama of a Squirrel And a Mad ...	174	NC-17

Thank You!

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